

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean code practices, and alignment with the defined requirements.

Instructions:

1. Code is well-structured, commented, and follows best practices for the Python and Flask technology stack.
2. The solution addresses the core problem statement and implements all key functional requirements.
3. All source files, configuration files, and setup instructions are included to run the application.
4. The code repository link is attached in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/1904Spandhana/hdi-prediction-ml
Programming Language(s)	Python (Data Science & Backend), HTML, CSS, JavaScript (Frontend & Dashboard)
Framework(s) Used	Flask (Web Framework), Scikit-learn (Machine Learning), Pandas & NumPy (Data Processing), Chart.js (Dashboard Visualizations)
Key Features Implemented	HDI prediction using Linear Regression; Flask web application with Home, Prediction Form and Result pages; Data preprocessing with null handling and feature selection; Model serialization with Pickle (HDI.pkl); 3 interactive analytics dashboards (HDI Score Comparison, World HDI Map, HDI Trend Over Years)
Pending / Incomplete Features	Production-grade error handling and input validation; User authentication; Cloud deployment of the Flask application; Real-time dataset updates from live UNDP API
Setup / Run Instructions	1) Clone the repository from GitHub. 2) Install dependencies: pip install numpy pandas matplotlib scikit-learn flask seaborn. 3) Navigate to the Flask folder. 4) Run the app using: python app.py. 5) Open http://127.0.0.1:5000 in a browser to use the application.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Prediction System successfully integrates a Linear Regression machine learning model with a Flask web application to deliver an accessible tool for predicting Human Development Index scores. The codebase follows a clean, modular folder structure separating Dataset, Flask, and Training components.

As an extension to the core requirements, three interactive analytics dashboards were added (HDI Score Comparison, World HDI Map, and HDI Trend Over Years), planned and led by Spandhana Mondam, with the dashboard coding implemented collaboratively by the remaining team members.